WORKED SOLUTION

STEP II 2008, Question 8

Pure • Vectors and Angle Bisectors

STEP II 2008, Question 8

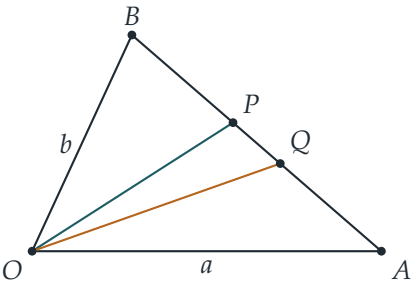
Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s2-q8

The key idea

A vector along the bisector of $\angle AOB$ is the sum of the *unit* vectors along OA and OB: adding two vectors of equal length gives the diagonal of a rhombus, and a rhombus diagonal bisects its angle. Matching this with the position vector of P fixes λ. After that, Q is just P with the roles of A and B swapped, and the final identity is a direct calculation with scalar products in which the awkward term $\mathbf{a}\cdot\mathbf{b}$ cancels.

Solution



In the picture $a > b$, so P (on the bisector, in teal) is nearer B, and Q is the point the same distance from B as P is from A.

The position vector of P

Since $AP : PB = (1-\lambda) : \lambda$ and the two parts add up to AB, we have $AP = (1-\lambda)AB$, so $\overrightarrow{AP} = (1-\lambda)(\mathbf{b}-\mathbf{a})$ and

$$\overrightarrow{OP} = \mathbf{a} + (1-\lambda)(\mathbf{b}-\mathbf{a}) = \lambda\mathbf{a} + (1-\lambda)\mathbf{b}.$$

(Check: $\lambda = 1$ gives A and $\lambda = 0$ gives B, as the ratio says.)

Finding λ

The vectors $\mathbf{a}/a$ and $\mathbf{b}/b$ are unit vectors along OA and OB. Their sum is the diagonal of a rhombus with sides along OA and OB, so it points along the bisector of $\angle AOB$. Since P lies on this bisector,

$$\lambda\mathbf{a} + (1-\lambda)\mathbf{b} = k\left(\frac{\mathbf{a}}{a} + \frac{\mathbf{b}}{b}\right)$$

for some scalar k.

Ask yourself Why may we compare the coefficients of $\mathbf{a}$ and of $\mathbf{b}$ separately?

Because O, A, B are not collinear, $\mathbf{a}$ and $\mathbf{b}$ are non-zero and not parallel, so $\mathbf{a}$ and $\mathbf{b}$ are linearly independent: if $\alpha\mathbf{a} + \beta\mathbf{b} = \mathbf{0}$ then $\alpha = \beta = 0$. Hence

$$\lambda = \frac{k}{a}, \qquad 1-\lambda = \frac{k}{b}.$$

Adding, $1 = k\left(\frac{1}{a} + \frac{1}{b}\right)$, so $k = \dfrac{ab}{a+b}$ and

$$\boxed{\lambda = \frac{b}{a+b}}, \qquad 1-\lambda = \frac{a}{a+b}.$$

This is the angle bisector theorem in disguise: $AP : PB = (1-\lambda) : \lambda = a : b = OA : OB$. So

$$\overrightarrow{OP} = \frac{b\,\mathbf{a} + a\,\mathbf{b}}{a+b}.$$

The point Q and the identity

Q is on AB with $BQ = AP = (1-\lambda)AB$, so by the same reasoning as for P, measured from B instead of from A,

$$\overrightarrow{OQ} = \mathbf{b} + (1-\lambda)(\mathbf{a}-\mathbf{b}) = (1-\lambda)\mathbf{a} + \lambda\mathbf{b} = \frac{a\,\mathbf{a} + b\,\mathbf{b}}{a+b}.$$

Write $c = \mathbf{a}\cdot\mathbf{b}$, and use $\mathbf{a}\cdot\mathbf{a} = a^2$, $\mathbf{b}\cdot\mathbf{b} = b^2$. Then

$$(a+b)^2\,OQ^2 = (a\mathbf{a} + b\mathbf{b})\cdot(a\mathbf{a} + b\mathbf{b}) = a^2\cdot a^2 + 2ab\,c + b^2\cdot b^2 = a^4 + b^4 + 2abc,$$
$$(a+b)^2\,OP^2 = (b\mathbf{a} + a\mathbf{b})\cdot(b\mathbf{a} + a\mathbf{b}) = b^2a^2 + 2ab\,c + a^2b^2 = 2a^2b^2 + 2abc.$$

Subtracting, the c terms cancel:

$$(a+b)^2\left(OQ^2 - OP^2\right) = a^4 - 2a^2b^2 + b^4 = (a^2-b^2)^2 = (a-b)^2(a+b)^2.$$

Dividing by $(a+b)^2 \neq 0$,

$$OQ^2 - OP^2 = (a-b)^2 = (b-a)^2,$$

as required.

Checks

Take $\mathbf{a} = (3,0)$ and $\mathbf{b} = (0,4)$, so $a = 3$, $b = 4$ and $\angle AOB$ is a right angle. Then $\lambda = \frac{4}{7}$ and $\overrightarrow{OP} = \left(\frac{12}{7}, \frac{12}{7}\right)$, which lies on $y = x$, the bisector of the right angle. Also $\overrightarrow{OQ} = \left(\frac{9}{7}, \frac{16}{7}\right)$, so $OQ^2 - OP^2 = \frac{337}{49} - \frac{288}{49} = 1 = (4-3)^2$. If $a = b$ the triangle is isosceles, $P = Q$ is the midpoint of AB, and the difference is 0, as it should be. A numerical test on 20,000 random triangles in two and three dimensions agreed with every result to within 10^{-13}.

Common slips

- Writing $\overrightarrow{OP} = (1-\lambda)\mathbf{a} + \lambda\mathbf{b}$: the larger share of the ratio, $1-\lambda$, is the distance from A, so it goes with $\mathbf{b}$. Test with $\lambda = 1$.
- Using $\mathbf{a} + \mathbf{b}$ as the bisector direction. That only works when $a = b$; you need the unit vectors.
- Expanding $|\mathbf{a} + \mathbf{b}|^2$ as $a^2 + b^2$, forgetting the cross term $2\,\mathbf{a} \cdot \mathbf{b}$. Here it cancels, but only if it is there in both expansions.

Beyond STEP

The final calculation is a special case of how lengths transform under a linear map: OP^2 and OQ^2 are values of the quadratic form $\mathbf{x} \mapsto \mathbf{x} \cdot \mathbf{x}$, and the cross terms are controlled by the Gram matrix of $\mathbf{a}$ and $\mathbf{b}$. The fact that unit vectors add to give a bisector also generalises: in any inner-product space, normalising and adding is how we find the direction "halfway" between two vectors. Strang's *Introduction to Linear Algebra* develops scalar products, lengths and angles in this spirit.

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